CODEFEST 3

1) Given

$N = 32$

FP32.

$C[i][j] = \sum A[i][k] * B[k][j]$

a) Each element of B accessed $= N^3$

$= (32)^3$

$= 32,768$

b) Total element accessed across A and B =

Elements across A + Elements across B

$= N^3 + N^3$

$= 32,768 + 32,768$

$= 65,536$

c) Total DRAM traffic = (Elements accessed in A and B + Elements accessed in c) × 4

$= (65,563 + (32)^2) \times 4$

$= 266,240$ bytes

2) tiled size $T = 8$

Matrix is divided: $\left(\frac{N}{T}\right)^2 = \left(\frac{32}{8}\right)^2 = 16$ tiles per matrix

Each tile holds $(T^2) = (8)^2 = 64$ elements

$= 64 \times 4$ bytes

$= 256$ bytes

2) a) DRAM loads for $A = N^2 \times 4$

$= (32)^2 \times 4$

$= 4096$ bytes.

b) Total DRAM loads for A and B

$= N^2 \times 4 \times 2$

$= 4096 \times 2$

$= 8192$ bytes.

b) DRAM traffic = total DRAM loads for A and B

$= 8192$ bytes

3) Ratio of DRAM traffic to tiled traffic

$$= \frac{\cancel{2} \times \cancel{N^3} \times \cancel{4}}{\cancel{2} \times \cancel{N^2} \times \cancel{4}} \qquad \frac{266,240}{8192}$$

$= 32$

If we take the approximate formula for total DRAM traffic naive method

$= 2 \times N^3 \times 4$

We are doing an approximation as output matrix access is negligible in comparison

The DRAM traffic formula for tiled method

$= 2 \times N^2 \times 4.$

Hence, the taking the ratio of the two, the value is N.

4) Naive Case:

Number of FLOPs needed $= 2 \times N^3$
$= 65,536$ FLOPs

Total DRAM traffic $= 2 \times N^3 \times 4$
$= 262,144$ bytes

Arithmetic Intensity $= \frac{65,536}{262,144}$
$= 0.25$ FLOP / byte.

Tiled Case:
Number of FLOPs needed $= 2N^3$
$= 65,536$ FLOPs

Total DRAM traffic $= 2 \times N^2 \times 4$
$= 8192$ bytes

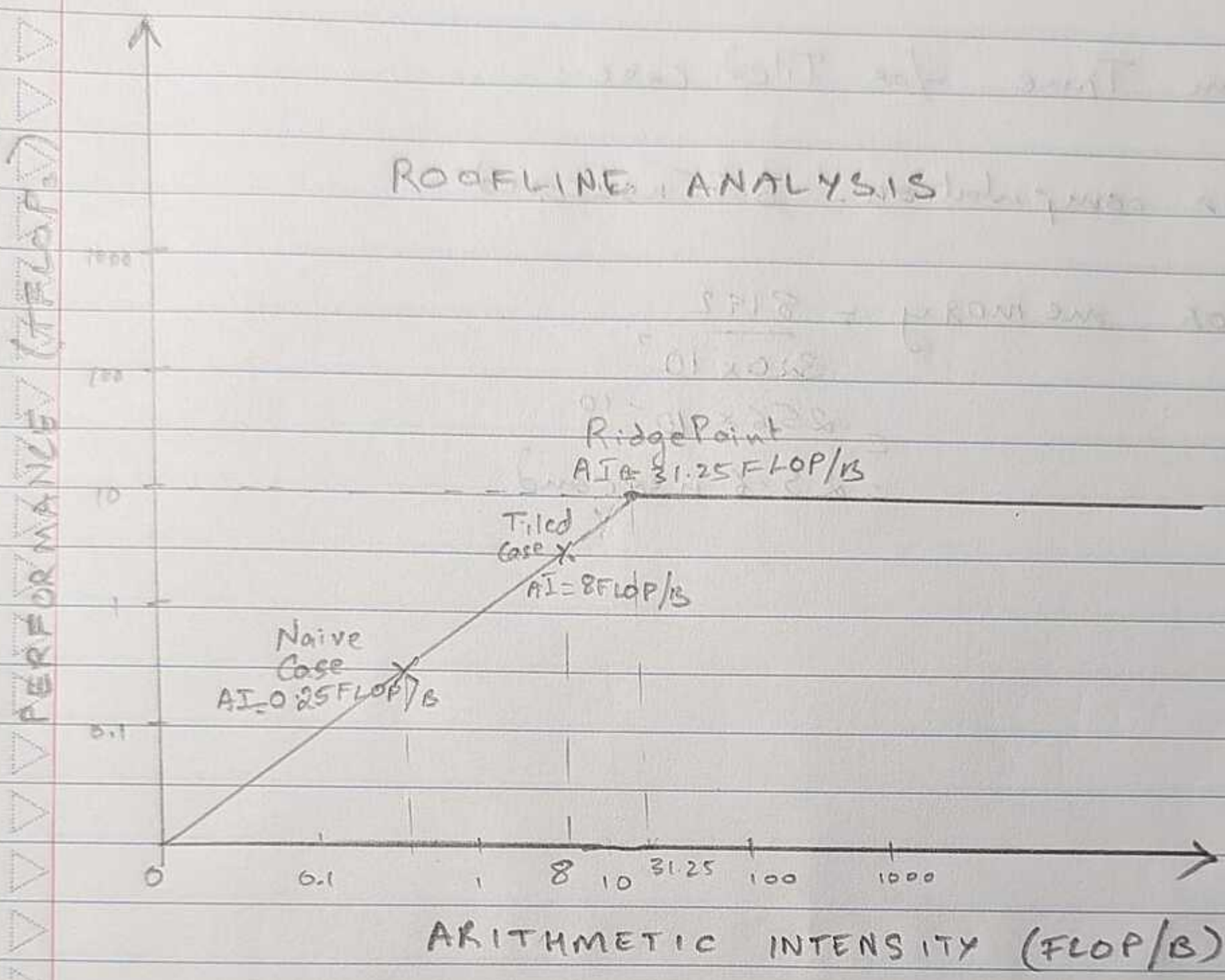
Arithmetic Intensity $= \frac{65,536}{8192}$

$= 8$ FLOPs / byte

Ridge Point:
B.W $= 320$ GB/s.
Peak Compute $= 10$ TFLOPs.

Ridge point $= \frac{\text{Peak compute}}{BW}$

$= \frac{10 \times 10^{12}}{320 \times 10^{9}} = 31.25$ FLOPs/byte



For both, the naive case and the tiled case, the bottleneck is memory bound.

Execution Time for Naive Case:

$$\text{time for computation} = \frac{\text{FLOPs needed}}{\text{Peak compute}}$$

$$= \frac{65{,}536}{10 \times 10^{12}} = 65536 \times 10^{-13} \text{ seconds.}$$

$$= 6.55 \text{ nseconds.}$$

$$\text{time for memory} = \frac{\text{Memory}}{\text{Peak BW}}$$

$$= \frac{262{,}144}{320 \times 10^{9}} = 8192 \times 10^{-10} \text{ seconds.}$$

$$= 819 \text{ nseconds.}$$

Execution Time for Tiled case:

time for computation = 6.55 nsecond

time for memory $= \frac{8192}{320 \times 10^7}$

$= 256 \times 10^{-10}$

$= 25.6$ nsecond